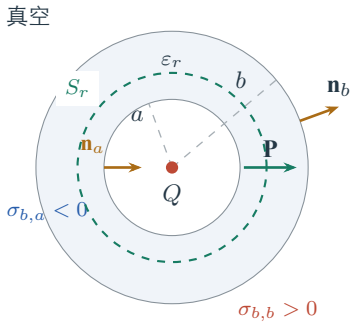
M3-010 介质球壳中的三种高斯定律



符号示意取 $Q > 0$, $\varepsilon_r > 1$

$$\Phi_D = Q\text{: 三个区域均相同}$$

$$a < r < b\text{: } q_b = -Q\alpha$$
$$\varepsilon_0 \Phi_E = Q/\varepsilon_r,\ \Phi_P = Q\alpha$$

$$r < a \text{ 或 } r > b\text{: } q_b = 0$$
$$\varepsilon_0 \Phi_E = Q,\ \Phi_P = 0$$